

Software Requirements Specification

Hospital Management System

A web application for patients, doctors, appointments, admissions and billing

Document information

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For academic use. Working product name: Carewell Hospital Management System.

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1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) describes the functional and non-functional requirements of the **Hospital Management System (HMS)**, a web-based application that helps a hospital manage patients, doctors, appointments, consultations, admissions, beds and billing in one place.

The document follows the structure recommended by IEEE 830-1998 and ISO/IEC/IEEE 29148:2018, scaled to suit a college software engineering project. It is the single source of truth for the development team, faculty evaluators, UI/UX designers, database and API designers, and testers.

1.2 Scope

The HMS is a role-based web application used by hospital staff and, in a limited way, by patients. It replaces paper registers and scattered spreadsheets with a connected system. Version 1.0 covers the following:

- Secure login with role-based access for Admin, Receptionist, Doctor and Patient users.
- Patient registration, search and record maintenance.
- Doctor and department management, including working days and shift timings.
- Appointment booking, rescheduling, cancellation and daily doctor schedules.
- Consultation notes, diagnosis and prescription records.
- Ward and bed management with patient admission and discharge.
- Invoice generation, payment recording and printable receipts.
- Dashboards and basic reports for hospital administration.

Items that are deliberately excluded from this version are listed in Section 2.7.

1.3 Intended audience and reading guide

Reader	Purpose	Recommended sections
Faculty and evaluators	Judge completeness, quality and feasibility of the requirements	All sections
Project team (developers)	Implement features and plan sprints	Sections 3, 4, 6, 8
UI/UX designers	Design screens and user flows	Sections 2.3, 3, 4, 5
Database and API designers	Create schema and endpoints	Sections 6 and 8
Testers	Write and run test cases	Sections 3, 7, 9

1.4 Definitions, acronyms and abbreviations

Term	Meaning
HMS	Hospital Management System, the product described in this document
SRS	Software Requirements Specification

Term	Meaning
RBAC	Role-Based Access Control, where permissions depend on the user role
OPD	Out-patient department, where patients are treated without being admitted
IPD	In-patient department, where patients are admitted to a ward or bed
UHID	Unique Health ID, the patient number generated by the HMS (for example P-1001)
CRUD	Create, Read, Update, Delete operations on a record
REST / API	Representational State Transfer; the interface used by the front end to talk to the server
UI / UX	User interface / user experience
FR / NFR	Functional requirement / non-functional requirement
UC / TC	Use case / test case
PII	Personally identifiable information, such as name, phone number and address

1.5 References

- IEEE Std 830-1998, Recommended Practice for Software Requirements Specifications.
- ISO/IEC/IEEE 29148:2018, Systems and software engineering: Life cycle processes: Requirements engineering.
- Digital Personal Data Protection Act, 2023 (India), for awareness of personal data handling duties.
- Information Technology Act, 2000 (India), for awareness of electronic records and security duties.
- OWASP Top 10, a reference list of common web application security risks.

1.6 Document overview

Section 2 gives the overall product description. Section 3 lists the functional requirements module by module. Section 4 covers external interfaces and the screen inventory. Section 5 contains the use case model, use case descriptions and the data flow diagram. Section 6 defines the data model and business rules. Section 7 lists the non-functional requirements. Section 8 proposes the technology stack, REST API plan and role permissions. Section 9 covers testing, acceptance criteria and requirements traceability. Section 10 gives the project plan, team roles and risks.

2. Overall Description

2.1 Product perspective

The HMS is a new, self-contained web application. It does not depend on any existing hospital software. Users reach it through a web browser. The browser talks to a server application through a REST API, and the server stores data in a relational database. Figure 1 shows the three-layer architecture.

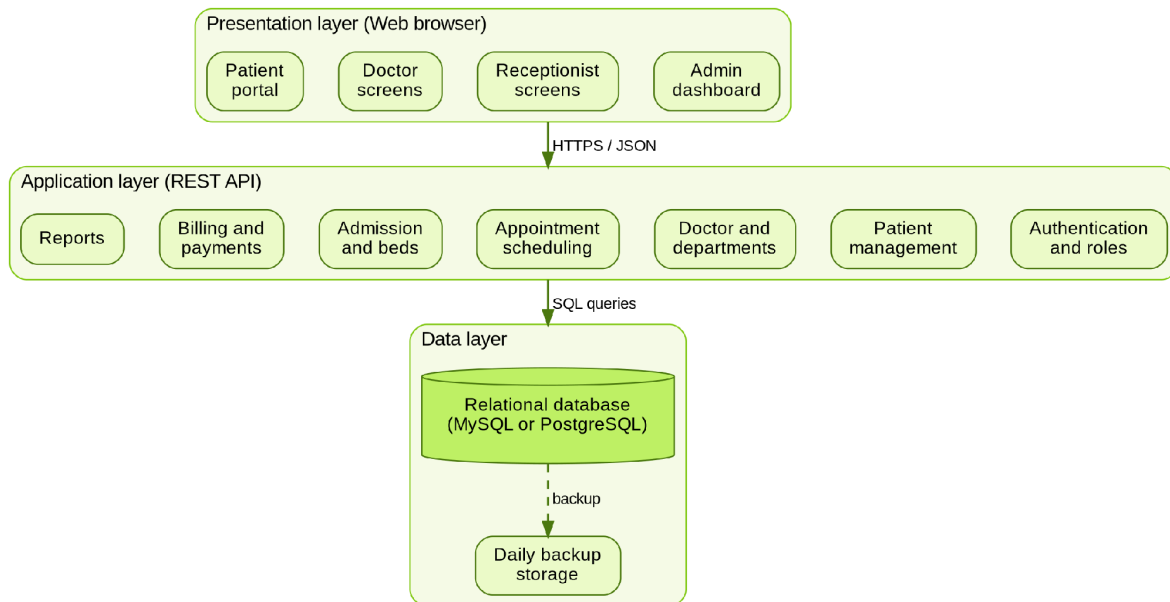


Figure 1: Three-layer architecture of the Hospital Management System

2.2 Product functions

Module	Summary of functions
Authentication and users	Login, logout, password change, role-based access, account management, audit log
Patient management	Register patients, generate UHID, search, edit, view visit and bill history
Doctor and department management	Maintain doctor profiles, departments, working days and shift timings
Appointment management	Book, reschedule, cancel and complete appointments; prevent double booking; daily schedule
Consultation and medical records	Record symptoms, diagnosis, prescription and follow-up advice
Admission and bed management	Maintain wards and beds, admit and discharge patients, show bed availability
Billing and payments	Create invoices from consultation and ward charges, record payments, print receipts
Dashboard and reports	Role-based dashboards, daily and monthly reports, export to CSV or PDF
Notifications	In-app alerts for appointments and dues; optional email reminders

2.3 User classes and characteristics

User class	Description	Technical skill	Main functions
Admin	Hospital administrator or IT officer who configures the system	Medium to high	Manage users, doctors, departments, beds, tariffs; view all reports; view audit logs
Receptionist	Front-desk staff who handle registration, booking and billing	Low to medium	Register patients, book appointments, admit and discharge, generate invoices, record payments
Doctor	Consultant or resident who treats patients	Low to medium	View own schedule, view patient history, record diagnosis and prescription, request admission
Patient	Registered patient using a self-service portal (limited access)	Low	View own profile, appointments, prescriptions and bills; request an appointment

2.4 Operating environment

- Client: any device with a modern browser (Chrome, Edge, Firefox or Safari, latest two versions), minimum screen width 360 px.
- Server: Windows, Linux or macOS machine, or a free-tier cloud host, able to run the chosen back-end runtime.
- Database: MySQL 8 or PostgreSQL 14 or later.
- Network: hospital LAN or internet connection with HTTPS enabled for any deployment outside localhost.

2.5 Design and implementation constraints

- The project must be completed within one academic semester by a team of four to five students.
- Only free and open-source tools and libraries may be used.
- The system must be a web application; native mobile apps are not required.
- Passwords must never be stored in plain text.
- All data entry screens must validate input on both the client and the server.
- The UI must use a consistent lime-green design theme and be usable on desktop and mobile screens.

2.6 Assumptions and dependencies

- Users have basic computer literacy and a working internet or LAN connection.
- The hospital has a single location and a single time zone (IST).
- All amounts are in Indian rupees (INR) and tax calculation is a flat, configurable percentage.
- Email delivery depends on a free SMTP service and is optional in version 1.0.
- Doctor timings and charges are supplied by the Admin before appointments can be booked.
- Sample data is used for demonstration; no real patient data is loaded during the project.

2.7 Out of scope and future enhancements

The following features are not part of version 1.0 but are good candidates for later phases:

- Pharmacy, inventory and pathology or radiology lab modules.
- Insurance and TPA claim processing, and online payment gateways.
- Integration with national health systems such as ABDM, and HL7 or FHIR data exchange.
- Native mobile apps, video consultation and wearable device integration.
- AI-assisted diagnosis, analytics dashboards and multi-branch hospital support.
- Biometric attendance and staff payroll.

2.8 Business objectives and success criteria

Objective	Success measure
Reduce patient registration time	A new patient can be registered in under 2 minutes
Avoid double booking of doctors	Zero overlapping appointments for a doctor in the same time slot
Give staff real-time bed visibility	Bed status on the dashboard matches actual admissions at all times
Reduce billing errors	Invoice totals are calculated automatically with no manual arithmetic
Protect patient data	Every page and API call is protected by login and role checks

3. System Features and Functional Requirements

This section lists what the system shall do. Each requirement has a unique ID in the form **FR-MODULE-NN**. The word "shall" marks a mandatory behaviour. Priority follows the MoSCoW idea in simple form:

- **High:** must be delivered in version 1.0; the system is not acceptable without it.
- **Medium:** should be delivered; important but a workaround exists.
- **Low:** could be delivered if time permits.

Tip for the team: build all High requirements first, then Medium, then Low. This gives a working system early and keeps the scope safe.

3.1 Authentication and user management

Description	Controls who can enter the system and what each person is allowed to do. It also records important actions for accountability.
Actors	Admin, Receptionist, Doctor, Patient
Related use cases	UC-01, UC-02
Related screens	SCR-01, SCR-15, SCR-16

ID	Requirement	Priority
FR-AUTH-01	The system shall let a user log in with a username (or email) and password.	High
FR-AUTH-02	The system shall lock an account for 15 minutes after 5 consecutive failed login attempts.	Medium
FR-AUTH-03	The system shall show only the menus, pages and actions allowed for the logged-in role (RBAC), and block direct access to other pages.	High
FR-AUTH-04	The system shall let a user log out, and shall end the session automatically after 30 minutes of inactivity.	High
FR-AUTH-05	The system shall let the Admin create, edit, deactivate and reactivate user accounts and assign a role to each.	High
FR-AUTH-06	The system shall require passwords of at least 8 characters with an uppercase letter, a lowercase letter and a digit, and shall store them only as salted hashes.	High
FR-AUTH-07	The system shall let users change their own password, and let the Admin reset a user's password.	Medium
FR-AUTH-08	The system shall record an audit log entry (user, action, record, date and time) for login, create, update, cancel and payment actions, viewable by the Admin.	Medium
FR-AUTH-09	The system shall let a Receptionist or Admin create a patient portal account linked to an existing patient record.	Medium
FR-AUTH-10	The system shall offer a forgot-password flow using an emailed one-time reset link.	Low

3.2 Patient management

Description	Maintains the master record of every patient, from first registration to complete visit history.
Actors	Receptionist, Admin, Doctor (read), Patient (own record, read)
Related use cases	UC-04, UC-05
Related screens	SCR-03, SCR-04, SCR-05

ID	Requirement	Priority
FR-PAT-01	The system shall let a Receptionist register a new patient with name, date of birth or age, gender, phone, blood group, address and emergency contact.	High
FR-PAT-02	The system shall generate a unique, sequential patient ID (UHID such as P-1001) when a patient is saved.	High
FR-PAT-03	The system shall warn the user when a new patient has the same name and phone number as an existing patient.	Medium
FR-PAT-04	The system shall let users search patients by UHID, name or phone number, including partial matches.	High
FR-PAT-05	The system shall let authorised users edit patient details and shall log every change.	High
FR-PAT-06	The system shall show a patient profile with demographics, appointments, medical records, admissions and invoices.	High
FR-PAT-07	The system shall let staff record known allergies and chronic conditions for a patient.	Medium
FR-PAT-08	The system shall allow a patient record to be deactivated but never permanently deleted when it has visit or billing history.	Medium
FR-PAT-09	The system shall show patient lists with pagination, sorting and filters for department and status.	Medium
FR-PAT-10	The system shall let staff print a patient registration card showing the UHID.	Low

3.3 Doctor and department management

Description	Keeps the list of doctors and departments and the timings that drive appointment booking.
Actors	Admin (manage), Receptionist and Patient (view)
Related use cases	UC-03
Related screens	SCR-06, SCR-07

ID	Requirement	Priority
FR-DOC-01	The system shall let the Admin add a doctor with name, department, specialization, qualification, phone, email and consultation fee.	High

ID	Requirement	Priority
FR-DOC-02	The system shall let the Admin edit or deactivate a doctor, and shall keep past appointments linked to that doctor.	High
FR-DOC-03	The system shall let the Admin add, edit and deactivate departments, and shall block deactivation of a department that still has active doctors.	High
FR-DOC-04	The system shall let the Admin set each doctor's available days, shift start and end time, and slot length (default 15 minutes).	High
FR-DOC-05	The system shall let the Admin mark a doctor on leave for a date range, and shall block bookings in that range.	Medium
FR-DOC-06	The system shall list doctors with search and filters by name, department and today's status (On duty, Off duty, On leave).	Medium
FR-DOC-07	The system shall show each doctor as a profile card with department, timings and fee.	Low

3.4 Appointment management

Description	Lets staff and patients book visits without clashes and gives doctors a clear daily schedule.
Actors	Receptionist, Patient, Doctor, Admin
Related use cases	UC-06, UC-07
Related screens	SCR-08, SCR-09

ID	Requirement	Priority
FR-APT-01	The system shall let a user book an appointment by choosing a patient, doctor, date, time slot and reason for visit.	High
FR-APT-02	The system shall show only free slots that fall inside the doctor's shift and outside leave periods.	High
FR-APT-03	The system shall prevent two active appointments for the same doctor, date and time slot, and shall show a clear message when a slot is taken.	High
FR-APT-04	The system shall not allow booking for a past date or time.	High
FR-APT-05	The system shall let authorised users reschedule or cancel an appointment before consultation, and shall require a reason for cancellation.	High
FR-APT-06	The system shall manage the appointment status through Scheduled, Checked-in, Completed, Cancelled and No-show.	High
FR-APT-07	The system shall show each doctor a daily schedule of appointments in time order.	High
FR-APT-08	The system shall let a Receptionist mark a patient as Checked-in on arrival and show a queue token number.	Medium
FR-APT-09	The system shall let a patient book and cancel own appointments from the portal under the same rules.	Medium

ID	Requirement	Priority
FR-APT-10	The system shall let users filter the appointment list by date, doctor, department, patient and status.	Medium
FR-APT-11	The system shall support same-day walk-in booking by suggesting the next free slot of the chosen doctor.	Medium

3.5 Consultation and medical records

Description	Captures what happens during a visit so that every doctor can see the patient's history.
Actors	Doctor (create and edit), Admin (read), Patient (own prescriptions, read)
Related use cases	UC-08
Related screens	SCR-10

ID	Requirement	Priority
FR-MED-01	The system shall let a Doctor record symptoms, vitals (blood pressure, pulse, temperature, weight), diagnosis and notes for an appointment.	High
FR-MED-02	The system shall let a Doctor add prescription lines with medicine, dose, frequency, duration and instructions.	High
FR-MED-03	The system shall show a patient's earlier records, newest first, while the Doctor is recording a consultation.	High
FR-MED-04	The system shall hide diagnosis and prescription details from the Receptionist role.	High
FR-MED-05	The system shall let the authoring Doctor edit a record within 24 hours; later changes shall be added as dated addenda.	Medium
FR-MED-06	The system shall let a Doctor set a follow-up date and shall show it on the patient profile.	Medium
FR-MED-07	The system shall provide a printable prescription with the hospital name, doctor, patient and date.	Medium
FR-MED-08	The system shall let a Doctor request admission with a reason, creating a pending admission request for the Receptionist.	Medium

3.6 Admission and bed management

Description	Tracks wards, beds and in-patients so staff always know which beds are free.
Actors	Receptionist (admit, discharge), Admin (setup), Doctor (request, read)
Related use cases	UC-09, UC-10, UC-15
Related screens	SCR-11

ID	Requirement	Priority
FR-ADM-01	The system shall let the Admin define wards and beds with bed number, bed type (General, Semi-private, Private, ICU) and daily tariff.	High
FR-ADM-02	The system shall track bed status as Available, Occupied, Reserved or Maintenance.	High
FR-ADM-03	The system shall let a Receptionist admit a patient by selecting the patient, bed, doctor, date and time, and reason.	High
FR-ADM-04	The system shall allow only Available beds to be assigned and shall allow only one active admission per patient.	High
FR-ADM-05	The system shall let a Receptionist discharge a patient with discharge date, time and summary, and shall set the bed back to Available or Maintenance.	High
FR-ADM-06	The system shall show a bed board that lists every bed by ward with a colour for each status.	High
FR-ADM-07	The system shall calculate the ward charge as number of days multiplied by the bed tariff, counting a minimum of one day.	High
FR-ADM-08	The system shall let a Receptionist transfer a patient to another available bed and record the transfer time.	Medium
FR-ADM-09	The system shall let users filter admissions by status (Admitted, Discharged) and by ward.	Medium

3.7 Billing and payments

Description	Turns consultations and hospital stays into accurate invoices and records how they are paid.
Actors	Receptionist (create, collect), Admin (approve, cancel), Patient (view)
Related use cases	UC-11, UC-12
Related screens	SCR-12, SCR-13

ID	Requirement	Priority
FR-BIL-01	The system shall create a consultation invoice automatically from the doctor's fee when an appointment is marked Completed.	High
FR-BIL-02	The system shall create an admission invoice from ward charges when a patient is discharged.	High
FR-BIL-03	The system shall let a Receptionist add extra line items (procedure, test, medicine, other) with description, quantity and unit price.	High
FR-BIL-04	The system shall calculate subtotal, discount, tax and total automatically.	High
FR-BIL-05	The system shall limit discounts to a configurable maximum percentage unless the Admin approves a higher value.	Medium
FR-BIL-06	The system shall record payments by cash, card or UPI with a reference number and shall support partial payments.	High

ID	Requirement	Priority
FR-BIL-07	The system shall manage invoice status as Unpaid, Partially paid, Paid or Cancelled and shall show the balance due.	High
FR-BIL-08	The system shall generate printable invoices and receipts with unique numbers (INV-0001, RCT-0001).	High
FR-BIL-09	The system shall not allow a Paid invoice to be edited; corrections shall be made by Admin cancellation with a reason and a new invoice.	High
FR-BIL-10	The system shall show a list of outstanding dues with search and filters.	Medium
FR-BIL-11	The system shall let a patient view own invoices and payment status in read-only mode.	Medium

3.8 Dashboard and reports

Description	Gives each role a quick summary and gives management the numbers needed for decisions.
Actors	Admin, Receptionist, Doctor, Patient
Related use cases	UC-13, UC-14
Related screens	SCR-02, SCR-14

ID	Requirement	Priority
FR-RPT-01	The system shall show the Admin a dashboard with total patients, today's appointments, doctors on duty, bed occupancy and today's revenue.	High
FR-RPT-02	The system shall show the Receptionist a dashboard with today's appointment queue, pending check-ins, new registrations and outstanding bills.	High
FR-RPT-03	The system shall show the Doctor a dashboard with today's appointment count, next patient and pending follow-ups.	Medium
FR-RPT-04	The system shall provide reports for daily registrations, appointments by doctor or department, revenue by date range, bed occupancy and outstanding dues.	Medium
FR-RPT-05	The system shall let users filter reports by date range, department and doctor.	Medium
FR-RPT-06	The system shall let users export a report to CSV and open a print-friendly view.	Medium
FR-RPT-07	The system shall show simple bar or line charts for revenue and appointment trends.	Low

3.9 Notifications

Description	Keeps users informed of events that need attention.
Actors	All roles
Related use cases	UC-06, UC-07
Related screens	SCR-02 (notification bell)

ID	Requirement	Priority
FR-NOT-01	The system shall create an in-app notification for the doctor and the patient when an appointment is booked, rescheduled or cancelled.	Medium
FR-NOT-02	The system shall show a notification bell with an unread count and let users mark notifications as read.	Medium
FR-NOT-03	The system shall alert the Receptionist when a patient is discharged with an unpaid invoice.	Low
FR-NOT-04	The system shall optionally send an email reminder 24 hours before an appointment.	Low

4. External Interface Requirements

4.1 User interface requirements

The user interface shall look like a modern, professional hospital application while staying simple to build with standard HTML, CSS and JavaScript. The following rules apply to all screens.

- **Layout:** a fixed sidebar with role-based menu items on the left and a content area on the right. On screens narrower than 760 px the sidebar collapses to a top menu.
- **Pages:** every list screen has search, sort and pagination; every form screen has labelled fields, inline validation and a single clear primary button such as "Save patient".
- **Wording:** plain, sentence-case labels. Buttons say exactly what happens ("Book appointment", not "Submit"). Error messages say what went wrong and how to fix it.
- **Accessibility:** text contrast of at least 4.5:1, visible keyboard focus, form labels linked to inputs, and no information conveyed by colour alone.
- **Feedback:** success and error messages appear after every save, delete or payment action. Destructive actions ask for confirmation.

Lime-green design theme

Role	Colour	Hex	Use
Primary	Lime green	#84CC16	Primary buttons, table headers, highlights
Primary dark	Forest lime	#4D7C0F	Links, active menu item, headings
Deep accent	Deep green	#365314	Sidebar, page titles, strong text on light lime
Surface	Pale lime	#ECFCCB / #F7FEE7	Cards, alternate table rows, form panels
Danger	Red	#DC2626	Errors, Cancelled, Occupied, Due
Warning	Amber	#F59E0B	Pending, Waiting, Partially paid
Neutral text	Charcoal	#1F2937	Body text on white backgrounds

4.2 Screen inventory

ID	Screen	Roles	Key elements
SCR-01	Login	All	Username, password, error message, forgot password link
SCR-02	Role dashboard	All	Summary cards, today's queue or schedule, notification bell
SCR-03	Patient list and search	Admin, Receptionist, Doctor	Search box, filters, patient table, Register patient button
SCR-04	Patient registration form	Receptionist, Admin	Demographic fields, blood group, emergency contact, duplicate warning
SCR-05	Patient profile	Admin, Receptionist, Doctor, Patient (own)	Tabs for details, appointments, records, admissions, bills

ID	Screen	Roles	Key elements
SCR-06	Doctor list	All staff, Patient (view)	Doctor cards, department filter, status tag
SCR-07	Doctor and department form	Admin	Doctor details, timings, leave dates, department list
SCR-08	Appointment booking	Receptionist, Patient	Patient picker, doctor, date, slot grid, reason
SCR-09	Appointment list and schedule	Receptionist, Doctor, Admin	Date filter, status tags, check-in, reschedule, cancel
SCR-10	Consultation	Doctor	Vitals, diagnosis, prescription table, history panel, follow-up date
SCR-11	Bed board and admissions	Receptionist, Admin, Doctor	Ward grid with bed colours, admit and discharge forms
SCR-12	Invoice	Receptionist, Admin, Patient (view)	Line items, discount, tax, total, status, print
SCR-13	Payment and receipt	Receptionist, Admin	Amount, method, reference, balance, receipt print
SCR-14	Reports	Admin, Receptionist (limited)	Report selector, filters, table, chart, export
SCR-15	User management	Admin	User table, role assignment, activate or deactivate, audit log
SCR-16	My account	All	Profile details, change password

4.3 Hardware interfaces

The HMS does not communicate directly with special hardware in version 1.0. Staff may use a standard office printer for invoices, receipts and prescriptions through the browser print function. Barcode or card readers are a future enhancement.

4.4 Software interfaces

Interface	Description
Web browser	Chrome, Edge, Firefox or Safari (latest two versions) running the HTML, CSS and JavaScript front end
Database server	MySQL 8 or PostgreSQL 14+, accessed only by the back end through parameterised queries
Email service (optional)	SMTP provider for password reset and appointment reminders
PDF generation library	Library used on the server or browser to create printable invoices and prescriptions

4.5 Communication interfaces

- All client-server communication shall use HTTP; deployments outside localhost shall use HTTPS (TLS 1.2 or higher).
- The front end and back end shall exchange data as JSON through the REST API described in Section 8.

- The API shall return standard status codes: 200 (OK), 201 (Created), 400 (Bad request), 401 (Not logged in), 403 (Not allowed), 404 (Not found) and 409 (Conflict, such as a double booking).

5. System Models

5.1 Use case model

Figure 2 shows which actors use which functions. The Admin and Receptionist are shown on the left, and the Doctor and Patient on the right.

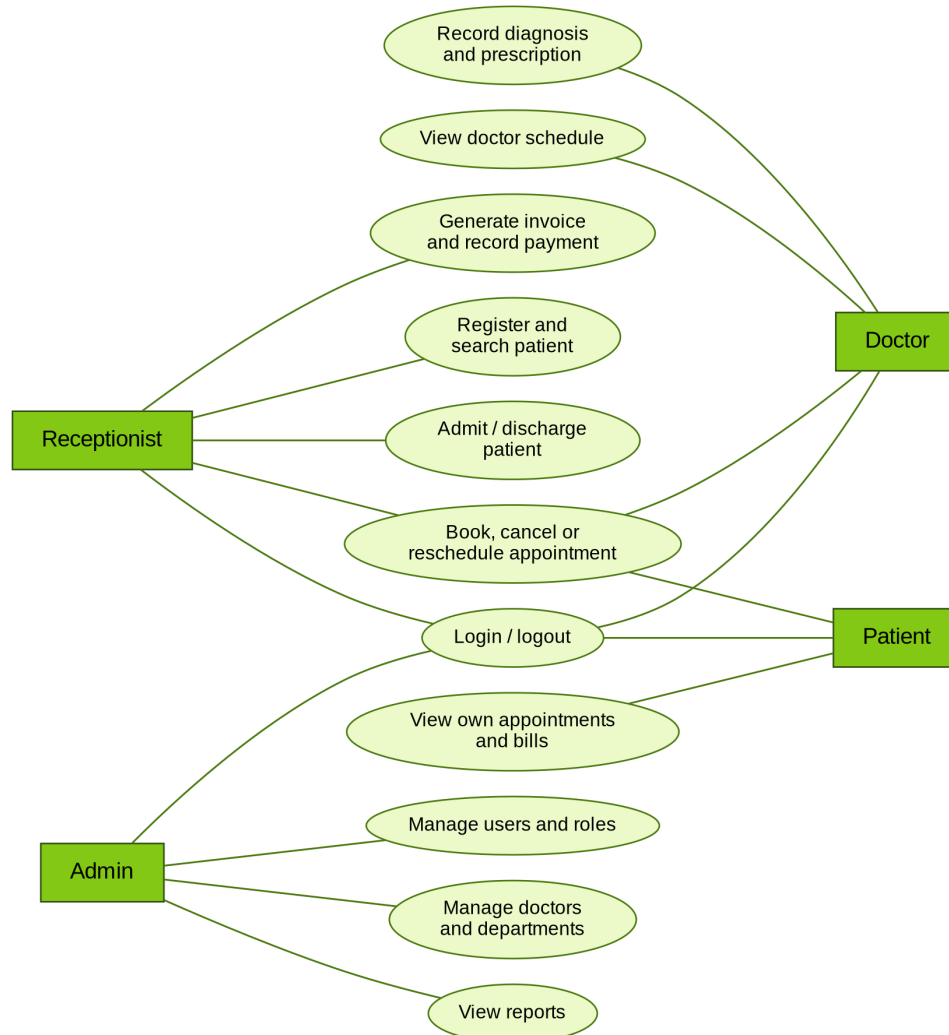


Figure 2: Use case diagram of the Hospital Management System

Use case summary

ID	Use case	Primary actor(s)	Priority
UC-01	Log in and log out	All users	High
UC-02	Manage users and roles	Admin	High
UC-03	Manage doctors and departments	Admin	High
UC-04	Register patient	Receptionist	High
UC-05	Search and view patient	Receptionist, Doctor, Admin	High
UC-06	Book appointment	Receptionist, Patient	High
UC-07	Reschedule or cancel appointment	Receptionist, Patient	High

ID	Use case	Primary actor(s)	Priority
UC-08	Record consultation	Doctor	High
UC-09	Admit patient	Receptionist	High
UC-10	Discharge patient	Receptionist	High
UC-11	Generate invoice	Receptionist	High
UC-12	Record payment	Receptionist	High
UC-13	View dashboard and reports	Admin, Receptionist, Doctor	Medium
UC-14	View own records (patient portal)	Patient	Medium
UC-15	Manage wards and beds	Admin	High

5.2 Detailed use case descriptions

The most important use cases are described below. The remaining use cases follow the same pattern and can be elaborated during design.

UC-01: Log in and log out

Actors	Admin, Receptionist, Doctor, Patient
Preconditions	The user has an active account.
Trigger	The user opens the HMS login page.
Main flow	1. User enters username and password. 2. System validates the credentials. 3. System creates a session and opens the dashboard for the user's role. 4. User works in the system and selects Log out when finished. 5. System ends the session and returns to the login page.
Alternative flows	2a. Credentials are wrong: system shows "Username or password is incorrect" and counts the failed attempt. 2b. Five failed attempts: system locks the account for 15 minutes.
Postconditions	A valid session exists (login) or no session exists (logout). The event is written to the audit log.
Related requirements	FR-AUTH-01 to FR-AUTH-04, FR-AUTH-08

UC-04: Register patient

Actor	Receptionist
Preconditions	Receptionist is logged in.
Trigger	A new patient arrives at the front desk.

Main flow	1. Receptionist selects Register patient. 2. System shows the registration form. 3. Receptionist enters name, date of birth or age, gender, phone, blood group, address and emergency contact. 4. Receptionist selects Save patient. 5. System validates the data and checks for a possible duplicate. 6. System generates the UHID, saves the record and shows the patient profile.
Alternative flows	5a. Mandatory field is missing or invalid: system highlights the field and explains the fix. 5b. Possible duplicate found: system shows the matching patient; Receptionist can open that record or continue saving.
Postconditions	A new patient with a unique UHID exists in the database.
Related requirements	FR-PAT-01 to FR-PAT-04

UC-06: Book appointment

Actors	Receptionist, Patient
Preconditions	User is logged in; the patient and doctor records exist; the doctor has timings set.
Trigger	A patient needs to see a doctor.
Main flow	1. User selects Book appointment. 2. User chooses the patient (Receptionist) and the doctor or department. 3. User picks a date; system shows the free slots for that doctor. 4. User selects a slot and enters the reason for the visit. 5. User selects Book appointment. 6. System re-checks the slot, saves the appointment with status Scheduled and shows a confirmation. 7. System sends in-app notifications to the doctor and the patient.
Alternative flows	3a. No free slots: system suggests the next available date. 6a. Slot was taken by another user: system shows "This slot was just booked. Please choose another" and refreshes the slots.
Postconditions	An appointment with status Scheduled exists and the slot is blocked.
Related requirements	FR-APT-01 to FR-APT-04, FR-NOT-01

UC-08: Record consultation

Actor	Doctor
Preconditions	Doctor is logged in; the appointment status is Checked-in.
Trigger	The patient enters the consultation room.

Main flow	1. Doctor opens the patient from the daily schedule. 2. System shows the patient's details and previous records. 3. Doctor enters vitals, symptoms, diagnosis and notes. 4. Doctor adds prescription lines and an optional follow-up date. 5. Doctor selects Save consultation. 6. System saves the record and marks the appointment Completed. 7. System creates the consultation invoice (FR-BIL-01).
Alternative flows	5a. Doctor needs the patient admitted: Doctor selects Request admission and enters a reason; system creates a pending admission request.
Postconditions	A medical record is stored; the appointment is Completed; an invoice exists.
Related requirements	FR-MED-01 to FR-MED-08, FR-BIL-01

UC-09: Admit patient

Actor	Receptionist
Preconditions	Patient exists and has no active admission; at least one bed is Available.
Trigger	A doctor requests admission or an emergency patient arrives.
Main flow	1. Receptionist opens the bed board and selects Admit patient. 2. Receptionist picks the patient, an Available bed, the treating doctor and the reason. 3. System checks that the bed is still Available and the patient has no active admission. 4. System creates the admission, sets the bed to Occupied and shows a confirmation.
Alternative flows	3a. Bed became occupied: system asks for another bed. 3b. Patient already admitted: system shows the existing admission.
Postconditions	Admission status is Admitted and the bed status is Occupied.
Related requirements	FR-ADM-03, FR-ADM-04

UC-11 / UC-12: Generate invoice and record payment

Actor	Receptionist
Preconditions	A completed appointment or a discharged admission exists.
Trigger	The patient is ready to pay.
Main flow	1. Receptionist opens the patient's invoice. 2. System shows the automatically created line items and total. 3. Receptionist adds extra items if needed and applies a discount within the allowed limit. 4. System recalculates the subtotal, tax and total. 5. Receptionist selects Record payment, enters amount, method and reference. 6. System saves the payment, updates the balance and invoice status. 7. Receptionist prints the invoice and receipt.
Alternative flows	3a. Discount is above the allowed limit: system asks for Admin approval. 5a. Amount is less than the balance: system records a partial payment and sets the status to Partially paid.
Postconditions	The payment is stored; invoice status is Partially paid or Paid; a receipt number exists.

Related requirements

FR-BIL-02 to FR-BIL-08

5.3 Context-level data flow diagram

Figure 3 shows the HMS as one process with the data exchanged between it and the outside actors, and the hospital database used to store the data.

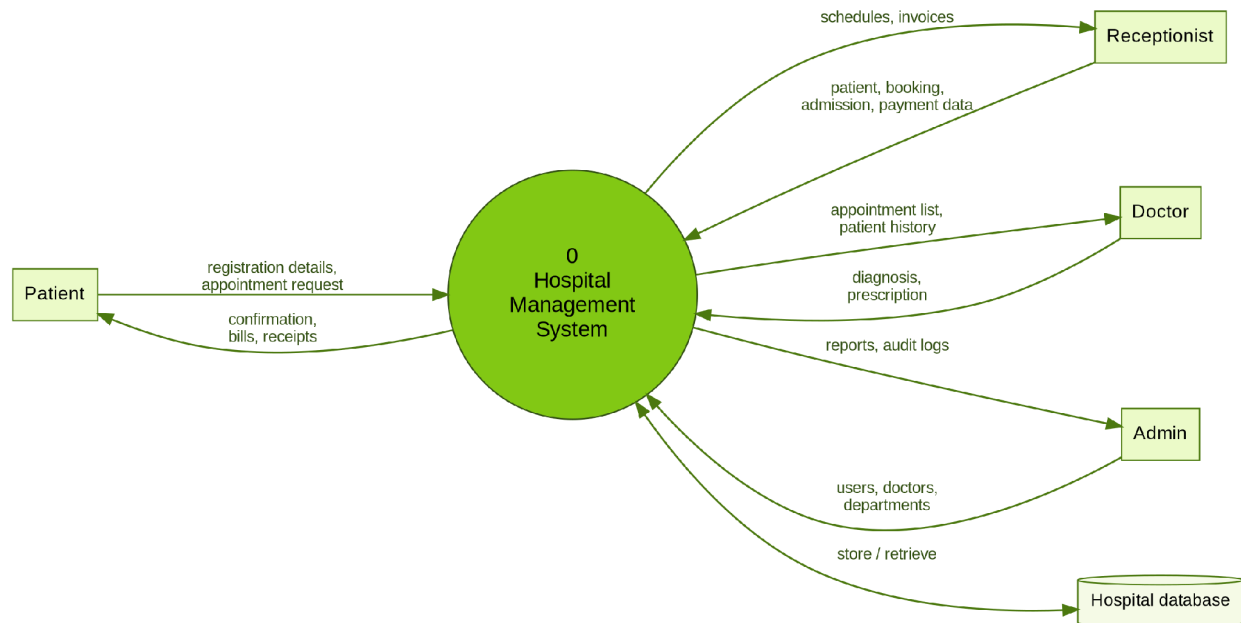


Figure 3: Level 0 data flow diagram (context diagram)

5.4 Appointment status lifecycle

From status	Event	To status	Allowed by
(new)	Appointment is booked	Scheduled	Receptionist, Patient
Scheduled	Date or time is changed	Scheduled	Receptionist, Patient
Scheduled	Patient arrives	Checked-in	Receptionist
Scheduled	Appointment is cancelled with reason	Cancelled	Receptionist, Patient, Admin
Scheduled	Slot time passes with no check-in	No-show	Receptionist (or system at end of day)
Checked-in	Consultation is saved	Completed	Doctor
Checked-in	Patient leaves before consultation	Cancelled	Receptionist

6. Data Requirements

6.1 Logical data model

Figure 4 shows the main entities and how they relate. A crow's foot at the end of a line means "many". For example, one doctor can have many appointments, and one invoice can have many payments. The diagram shows the key attributes only; the full field list is in Section 6.2.

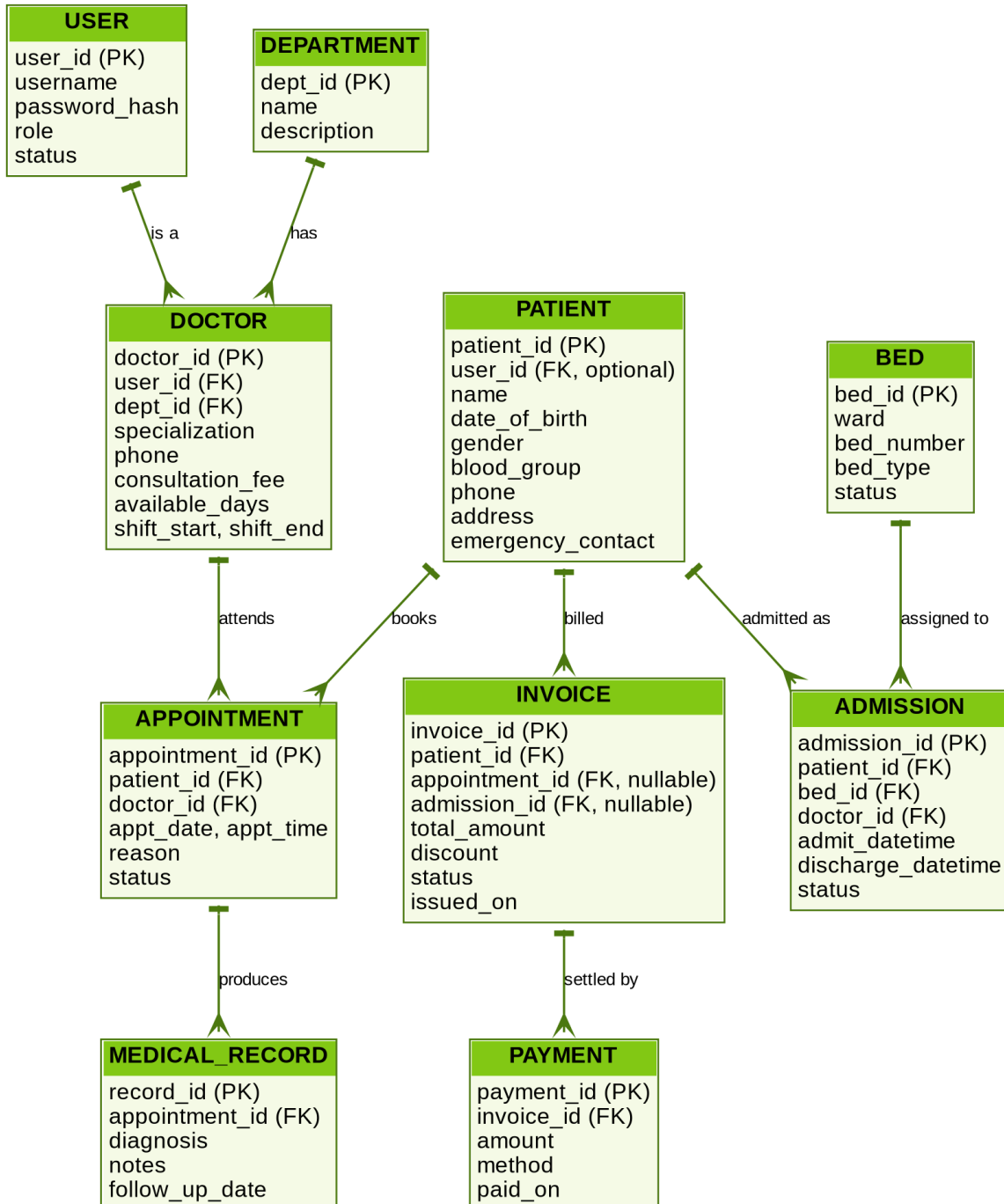


Figure 4: Entity relationship diagram of the Hospital Management System

6.2 Data dictionary

Data types use MySQL-style names for clarity. They can be adapted for PostgreSQL. Every table also has created_at and updated_at timestamp columns.

USER

Field	Type	Constraints	Description
user_id	INT	PK, auto-increment	Unique account number
username	VARCHAR(50)	UNIQUE, NOT NULL	Login name
email	VARCHAR(100)	UNIQUE	Used for reset links and reminders
password_hash	VARCHAR(255)	NOT NULL	Salted hash, never plain text
role	ENUM	NOT NULL	Admin, Receptionist, Doctor, Patient
status	ENUM	NOT NULL, default Active	Active, Inactive, Locked
failed_attempts	INT	default 0	Consecutive failed logins
last_login	DATETIME	NULL	Time of last successful login

PATIENT

Field	Type	Constraints	Description
patient_id	INT	PK, auto-increment	Internal key, displayed as UHID (P-1001)
user_id	INT	FK to USER, NULL	Portal account, if any
name	VARCHAR(100)	NOT NULL	Full name
date_of_birth	DATE	NOT NULL	Used to calculate age
gender	ENUM	NOT NULL	Female, Male, Other
blood_group	ENUM	NULL	A+, A-, B+, B-, AB+, AB-, O+, O-
phone	VARCHAR(10)	NOT NULL	10-digit mobile number
address	VARCHAR(255)	NULL	Street and city
emergency_contact	VARCHAR(100)	NULL	Name and phone of a relative
allergies	TEXT	NULL	Known allergies and chronic conditions
status	ENUM	default Active	Active or Inactive

DEPARTMENT

Field	Type	Constraints	Description
dept_id	INT	PK, auto-increment	Unique department number
name	VARCHAR(80)	UNIQUE, NOT NULL	For example Cardiology
description	VARCHAR(255)	NULL	Short description
status	ENUM	default Active	Active or Inactive

DOCTOR

Field	Type	Constraints	Description
doctor_id	INT	PK, auto-increment	Unique doctor number
user_id	INT	FK to USER, UNIQUE	Login account of the doctor
dept_id	INT	FK to DEPARTMENT	Department the doctor works in
full_name	VARCHAR(100)	NOT NULL	Name shown on screens and prescriptions
specialization	VARCHAR(100)	NOT NULL	For example Interventional cardiology
qualification	VARCHAR(100)	NULL	For example MBBS, MD
phone	VARCHAR(10)	NOT NULL	Contact number
consultation_fee	DECIMAL(10,2)	NOT NULL, at least 0	Fee used for the consultation invoice
available_days	VARCHAR(30)	NOT NULL	For example Mon,Tue,Wed,Thu,Fri
shift_start, shift_end	TIME	NOT NULL, start before end	Working hours for slot generation
slot_minutes	INT	default 15	Length of one appointment slot
status	ENUM	default Active	Active or Inactive

APPOINTMENT

Field	Type	Constraints	Description
appointment_id	INT	PK, auto-increment	Unique appointment number
patient_id	INT	FK to PATIENT	Who is visiting
doctor_id	INT	FK to DOCTOR	Whom they visit
appt_date	DATE	NOT NULL	Date of visit
appt_time	TIME	NOT NULL	Start of slot
reason	VARCHAR(255)	NULL	Reason for visit
status	ENUM	NOT NULL	Scheduled, Checked-in, Completed, Cancelled, No-show
token_no	INT	NULL	Queue number given at check-in
cancel_reason	VARCHAR(255)	NULL	Required when status is Cancelled
Unique rule	-	(doctor_id, appt_date, appt_time) for active statuses	Prevents double booking

MEDICAL_RECORD

Field	Type	Constraints	Description
record_id	INT	PK, auto-increment	Unique record number

Field	Type	Constraints	Description
appointment_id	INT	FK to APPOINTMENT, UNIQUE	Visit this record belongs to
symptoms	TEXT	NULL	Symptoms reported
bp, pulse, temperature, weight	VARCHAR / INT / DECIMAL	NULL	Vitals measured during the visit
diagnosis	TEXT	NOT NULL	Doctor's diagnosis
notes	TEXT	NULL	Additional advice
follow_up_date	DATE	NULL	Suggested date of next visit

BED

Field	Type	Constraints	Description
bed_id	INT	PK, auto-increment	Unique bed number
ward	VARCHAR(50)	NOT NULL	For example General ward A
bed_number	VARCHAR(10)	NOT NULL	Number within the ward
bed_type	ENUM	NOT NULL	General, Semi-private, Private, ICU
daily_tariff	DECIMAL(10,2)	NOT NULL	Charge per day
status	ENUM	default Available	Available, Occupied, Reserved, Maintenance
Unique rule	-	(ward, bed_number)	No duplicate beds in a ward

ADMISSION

Field	Type	Constraints	Description
admission_id	INT	PK, auto-increment	Unique admission number
patient_id	INT	FK to PATIENT	Admitted patient
bed_id	INT	FK to BED	Current bed
doctor_id	INT	FK to DOCTOR	Treating doctor
admit_datetime	DATETIME	NOT NULL	Admission time
discharge_datetime	DATETIME	NULL	Set at discharge
reason	VARCHAR(255)	NOT NULL	Reason for admission
discharge_summary	TEXT	NULL	Summary written at discharge
status	ENUM	NOT NULL	Admitted or Discharged

INVOICE

Field	Type	Constraints	Description
invoice_id	INT	PK, auto-increment	Internal key

Field	Type	Constraints	Description
invoice_no	VARCHAR(12)	UNIQUE, NOT NULL	Printed number such as INV-0001
patient_id	INT	FK to PATIENT	Billed patient
appointment_id	INT	FK, NULL	Set for consultation invoices
admission_id	INT	FK, NULL	Set for admission invoices
subtotal, discount, tax	DECIMAL(10,2)	at least 0	Amounts before and after adjustment
total_amount	DECIMAL(10,2)	NOT NULL	Final payable amount
status	ENUM	NOT NULL	Unpaid, Partially paid, Paid, Cancelled
issued_on	DATETIME	NOT NULL	Invoice creation time

PAYMENT

Field	Type	Constraints	Description
payment_id	INT	PK, auto-increment	Internal key
receipt_no	VARCHAR(12)	UNIQUE, NOT NULL	Printed number such as RCT-0001
invoice_id	INT	FK to INVOICE	Invoice being paid
amount	DECIMAL(10,2)	greater than 0	Amount received
method	ENUM	NOT NULL	Cash, Card, UPI
reference_no	VARCHAR(50)	NULL	Card or UPI transaction reference
paid_on	DATETIME	NOT NULL	Payment time
received_by	INT	FK to USER	Staff member who took the payment

Supporting tables

Table	Purpose	Key fields
DOCTOR_LEAVE	Leave periods that block booking	leave_id, doctor_id, from_date, to_date, reason
PRESCRIPTION_ITEM	Medicine lines of a prescription	item_id, record_id, medicine, dose, frequency, duration, instructions
INVOICE_ITEM	Line items of an invoice	item_id, invoice_id, description, quantity, unit_price, amount
NOTIFICATION	In-app notifications	notification_id, user_id, message, is_read, created_at
AUDIT_LOG	Record of important actions	log_id, user_id, action, entity, entity_id, created_at
SETTINGS	Configurable values	key, value (tax percent, maximum discount percent, default slot length)

6.3 Business rules

ID	Rule
BR-01	Every patient has exactly one UHID, and a UHID is never reused.
BR-02	A doctor cannot have two active appointments on the same date and time.
BR-03	An appointment can be booked only for today or a later date, inside the doctor's shift, and not on a leave day.
BR-04	A patient can have only one active admission at a time.
BR-05	Only an Available bed can be assigned. The bed becomes Occupied at admission and Available or Maintenance at discharge.
BR-06	Ward charge equals the number of calendar days between admission and discharge dates (minimum 1) multiplied by the bed's daily tariff.
BR-07	Invoice total equals the sum of line items, minus discount, plus tax calculated on the discounted amount.
BR-08	A discount above 20% of the subtotal needs Admin approval. The limit is configurable.
BR-09	A payment must be greater than zero and not more than the balance due. The invoice becomes Paid when the balance reaches zero.
BR-10	A Paid invoice cannot be edited. Only the Admin can cancel it, with a reason.
BR-11	A doctor can edit only own medical records, and only within 24 hours of saving. Later changes are addenda.
BR-12	The Receptionist role cannot see diagnosis or prescription details.
BR-13	A patient user can see only own profile, appointments, prescriptions and invoices.
BR-14	Users, doctors, departments and patients with history are deactivated, never deleted.

6.4 Input validation rules

Field	Validation rule
Name	2 to 100 characters; letters, spaces, dots and hyphens only
Phone	Exactly 10 digits, starting with 6, 7, 8 or 9
Email	Valid email format, maximum 100 characters
Date of birth	Not in the future; age not above 120 years
Appointment date and time	Not in the past; must match a free slot
Amounts and fees	Positive number with at most 2 decimal places
Quantity	Whole number of 1 or more
Password	At least 8 characters with uppercase, lowercase and a digit
Free-text fields	Trimmed, length-limited and encoded on output to prevent script injection

7. Non-Functional Requirements

Non-functional requirements describe how well the system must work. Each has an ID in the form **NFR-AREA-NN** and a priority.

7.1 Performance

ID	Requirement	Priority
NFR-PERF-01	Pages shall load within 3 seconds on a standard broadband or hospital LAN connection.	High
NFR-PERF-02	Patient search shall return results within 2 seconds for up to 10,000 patient records.	Medium
NFR-PERF-03	Saving a form (registration, booking, payment) shall complete within 2 seconds.	Medium
NFR-PERF-04	A one-month report shall be generated within 10 seconds.	Medium
NFR-PERF-05	The system shall support at least 50 concurrent users in the demonstration environment without errors.	Medium

7.2 Security

ID	Requirement	Priority
NFR-SEC-01	Every page and API endpoint except login shall require authentication.	High
NFR-SEC-02	Passwords shall be hashed with a salted algorithm such as bcrypt or Argon2.	High
NFR-SEC-03	Role permissions shall be enforced on the server for every request, not only hidden in the interface.	High
NFR-SEC-04	The system shall protect against SQL injection (parameterised queries), cross-site scripting (output encoding) and cross-site request forgery (tokens or same-site cookies).	High
NFR-SEC-05	Deployments outside localhost shall use HTTPS, and session cookies shall be HttpOnly and Secure.	High
NFR-SEC-06	Diagnosis and prescription data shall be visible only to authorised roles, and patients shall see only their own data.	High
NFR-SEC-07	Sessions shall expire after 30 minutes of inactivity.	High
NFR-SEC-08	Audit log entries shall not be editable or deletable from the user interface.	Medium
NFR-SEC-09	Passwords, keys and database credentials shall be kept in environment variables, not in source code.	Medium

7.3 Usability

ID	Requirement	Priority
NFR-USE-01	The layout, terms and colours shall be consistent across all screens.	High
NFR-USE-02	The interface shall work from 360 px to 1920 px screen width.	High
NFR-USE-03	A new staff member shall be able to register a patient, book an appointment and record a payment after 30 minutes of training.	Medium
NFR-USE-04	Frequent tasks (register patient, book appointment, record payment) shall be reachable within 3 clicks from the dashboard.	Medium
NFR-USE-05	Text contrast shall be at least 4.5:1, all controls shall be keyboard accessible, and form fields shall have labels.	Medium
NFR-USE-06	Error messages shall explain the problem and the fix in plain language.	Medium

7.4 Reliability and availability

ID	Requirement	Priority
NFR-REL-01	Foreign keys and constraints shall protect data integrity in the database.	High
NFR-REL-02	Booking, admission and payment actions shall be atomic: either all steps are saved or none.	High
NFR-REL-03	The database shall be backed up daily and a restore shall be tested at least once before final submission.	High
NFR-REL-04	The system shall be available at least 99% of the time during hospital hours (8 am to 8 pm).	Medium
NFR-REL-05	Users shall never see technical error traces; errors shall be logged on the server.	Medium

7.5 Maintainability, portability and scalability

ID	Requirement	Priority
NFR-MNT-01	The code shall be organised by feature with consistent naming and comments for complex logic.	Medium
NFR-MNT-02	A README shall explain setup, environment variables, database scripts and sample data.	High
NFR-MNT-03	The source code shall be in Git, and every team member shall commit regularly with meaningful messages.	Medium
NFR-MNT-04	Tax percentage, maximum discount and default slot length shall be configurable and not hard-coded.	Medium
NFR-MNT-05	Business rules BR-01 to BR-10 shall have automated unit tests where practical.	Low
NFR-MNT-06	The system shall run on Windows, Linux and macOS and on the latest two versions of Chrome, Edge, Firefox and Safari without plugins.	High

ID	Requirement	Priority
NFR-MNT-07	The database shall use indexes on search fields so it can grow to 100,000 patients and 500 appointments per day.	Low

7.6 Legal and ethical

ID	Requirement	Priority
NFR-LEG-01	The system shall collect only the personal data needed for care and billing.	Medium
NFR-LEG-02	A short privacy notice shall be shown on the login page.	Low
NFR-LEG-03	Records shall be kept for a retention period set by hospital policy (project default: 5 years) and shall not be erased earlier.	Low
NFR-LEG-04	Demonstration and test data shall be fictional; no real patient data shall be used in the project.	High

8. Technology Stack, API Plan and Access Control

8.1 Suggested technology stack

The stack below is a suggestion that suits a student team. The team may choose another option if it covers the same needs.

Layer	Recommended option	Alternatives
Front end	HTML5, CSS3, JavaScript (ES6) with a lime-green theme	React with Vite; Bootstrap or Tailwind CSS for faster styling
Back end	Node.js with Express	Java Spring Boot, Python Django or Flask, PHP Laravel
Database	MySQL 8	PostgreSQL 14+
Authentication	Session cookie or JWT with bcrypt password hashing	OAuth is not required
API testing	Postman collection	Insomnia, Jest with Supertest
Design and diagrams	Figma, draw.io	Canva, Lucidchart
Version control	Git with GitHub	GitLab
Hosting	Local machine for demo; free-tier cloud host if needed	Any VPS or university server

8.2 REST API plan

All endpoints return JSON. List endpoints accept page and size parameters. Protected endpoints require a valid session. The table is a plan for a later design phase; paths may be refined by the team.

Method	Endpoint	Purpose	Roles
POST	/api/auth/login	Log in	All
POST	/api/auth/logout	Log out	All
PUT	/api/auth/password	Change own password	All
GET, POST	/api/users	List or create users	Admin
PUT	/api/users/{id}	Edit, deactivate or reset a user	Admin
GET	/api/patients?q=	Search patients	Admin, Receptionist, Doctor
POST	/api/patients	Register a patient	Receptionist, Admin
GET	/api/patients/{id}	Patient profile	Staff; Patient (own)
PUT	/api/patients/{id}	Update a patient	Receptionist, Admin
GET, POST	/api/departments	List or add departments	All (GET); Admin (POST)
GET, POST	/api/doctors	List or add doctors	All (GET); Admin (POST)
PUT	/api/doctors/{id}	Edit doctor, timings or leave	Admin

Method	Endpoint	Purpose	Roles
GET	/api/doctors/{id}/slots?date=	Free slots for a date	Receptionist, Patient
POST	/api/appointments	Book an appointment	Receptionist, Patient
GET	/api/appointments?date=&doctor=	List or filter appointments	Receptionist, Doctor, Admin
PUT	/api/appointments/{id}	Reschedule	Receptionist, Patient
PATCH	/api/appointments/{id}/status	Check in, cancel, no-show	Receptionist, Patient (cancel)
POST	/api/appointments/{id}/record	Save consultation and prescription	Doctor
GET	/api/patients/{id}/records	Medical history	Doctor, Admin; Patient (own)
GET	/api/beds	Bed board	Receptionist, Admin, Doctor
POST	/api/admissions	Admit a patient	Receptionist
PATCH	/api/admissions/{id}/discharge	Discharge a patient	Receptionist
GET	/api/invoices/{id}	View an invoice	Receptionist, Admin; Patient (own)
POST	/api/invoices/{id}/items	Add an extra line item	Receptionist
POST	/api/invoices/{id}/payments	Record a payment	Receptionist
GET	/api/dashboard	Dashboard data for the role	All
GET	/api/reports/{type}	Run a report	Admin, Receptionist
GET	/api/notifications	List notifications	All
GET	/api/audit-logs	View audit log	Admin

8.3 Role-permission matrix

Function	Admin	Receptionist	Doctor	Patient
Manage users and roles	Yes	-	-	-
Manage doctors and departments	Yes	View	View	View
Manage wards and beds	Yes	View	View	-
Register and edit patients	Yes	Yes	-	-
View patient demographics	Yes	Yes	Yes	Own
Book and cancel appointments	Yes	Yes	-	Own
View doctor schedule	Yes	Yes	Own	-
Record consultation and prescription	-	-	Yes	-
View diagnosis and prescriptions	Yes	-	Yes	Own
Admit and discharge patients	Yes	Yes	Request	-
Create invoices and record payments	Yes	Yes	-	-
Approve discounts, cancel invoices	Yes	-	-	-

Function	Admin	Receptionist	Doctor	Patient
View invoices	Yes	Yes	-	Own
View reports	Yes	Limited	-	-
View audit log	Yes	-	-	-

9. Testing, Acceptance and Traceability

9.1 Test strategy

Test level	Focus	Done by	Tools
Unit testing	Business rules, validations and calculations (for example invoice totals, slot generation)	Developers	Jest, JUnit or PyTest
Integration testing	API endpoints working with the database	Developers, testers	Postman collection
System testing	Complete workflows across modules, such as register, book, consult, bill and pay	Tester	Written test scripts, manual execution
Security testing	Login, role access, injection and session expiry	Tester	Manual checks, OWASP ZAP (optional)
Usability and compatibility	Screen sizes from 360 px, browser support, accessibility checks	UI designer, tester	Browser developer tools
User acceptance testing	Faculty or mock users confirm the criteria in Section 9.2	Whole team with faculty	Acceptance checklist

9.2 Acceptance criteria

The system is accepted when the criteria below are demonstrated with sample data.

Module	Acceptance criteria
Authentication	Each of the four roles can log in and sees only its own menu. A wrong password is rejected. A session ends after inactivity.
Patients	A new patient is saved with a UHID in under 2 minutes. Search by name, phone and UHID finds the patient.
Doctors	Admin can add a doctor with timings; the booking screen shows correct slots for that doctor only.
Appointments	A taken slot cannot be booked again. Cancelled slots become free. The doctor sees today's schedule in time order.
Consultation	A doctor saves diagnosis and prescription; the appointment becomes Completed and an invoice is created.
Admission	Occupied beds cannot be assigned. Discharge frees the bed and calculates the ward charge correctly.
Billing	Totals, discount and tax are calculated automatically. Partial payment updates the balance. A Paid invoice cannot be edited.
Reports	Dashboard numbers match the data in the database. A report can be filtered and exported to CSV.
Security	Direct URL access to a page of another role is blocked. Passwords are stored as hashes.

9.3 Sample test cases

These cases cover the most important requirements. The team shall extend them into a full test plan during the testing phase.

TC ID	Requirement	Scenario	Expected result
TC-001	FR-AUTH-01	Log in with valid credentials as each role	The correct role dashboard opens
TC-002	FR-AUTH-01	Log in with a wrong password	Error message appears and no session is created
TC-003	FR-AUTH-02	Enter a wrong password 5 times	Account is locked for 15 minutes
TC-004	FR-AUTH-03	Receptionist opens the user management URL directly	Access denied (403) message
TC-005	FR-AUTH-04	Stay idle for 30 minutes	User is redirected to the login page
TC-006	FR-PAT-01, 02	Register a patient with valid data	Patient is saved with a unique UHID
TC-007	FR-PAT-01	Register a patient with phone number 12345	Field error: phone must be 10 digits
TC-008	FR-PAT-03	Register a patient with an existing name and phone	Duplicate warning is shown
TC-009	FR-PAT-04	Search for part of a patient's name	Matching patients are listed
TC-010	FR-DOC-04	Open booking for a doctor with shift 9:00 to 13:00 and 15-minute slots	16 slots are shown
TC-011	FR-APT-03	Book a slot that is already booked	Conflict message; no new appointment is created
TC-012	FR-APT-04	Book an appointment for yesterday	Booking is blocked with a clear message
TC-013	FR-APT-05	Cancel an appointment without a reason	Reason is required before cancellation
TC-014	FR-DOC-05	Open booking for a doctor on a leave day	No slots are shown for that date
TC-015	FR-MED-01, FR-BIL-01	Doctor saves a consultation	Appointment becomes Completed and an invoice is created
TC-016	FR-MED-04	Receptionist opens a patient's consultation record	Diagnosis and prescription are hidden
TC-017	FR-MED-05	Doctor edits a record after 24 hours	Direct edit is blocked; only an addendum can be added
TC-018	FR-ADM-04	Admit a patient to an Occupied bed	Action is blocked; another bed must be chosen
TC-019	FR-ADM-05	Discharge an admitted patient	Admission is Discharged and the bed is Available
TC-020	FR-ADM-07	Admit on 1 October and discharge on 4 October at Rs 1,500 per day	Ward charge is Rs 4,500
TC-021	FR-BIL-04	Invoice with subtotal Rs 2,000, discount 10% and tax 5%	Total is Rs 1,890
TC-022	FR-BIL-05	Receptionist applies a 30% discount	Admin approval is required
TC-023	FR-BIL-06, 07	Pay Rs 500 against an invoice of Rs 1,890	Status is Partially paid and balance is Rs 1,390

TC ID	Requirement	Scenario	Expected result
TC-024	FR-BIL-09	Edit a Paid invoice	Edit is blocked; only Admin cancellation is possible
TC-025	FR-RPT-01	Compare Admin dashboard figures with database counts	Numbers match
TC-026	NFR-SEC-04	Type an SQL injection string into the patient search	Text is treated as plain input; no error or data leak
TC-027	NFR-USE-02	Open all main screens at 360 px width	No horizontal scrolling; all controls usable
TC-028	NFR-PERF-02	Search with 10,000 patient records loaded	Results appear within 2 seconds

9.4 Requirements traceability matrix

The matrix links each module to its requirements, use cases, screens and test cases so that nothing is built or tested without a reason.

Module	Requirements	Use cases	Screens	Test cases
Authentication	FR-AUTH-01 to 10	UC-01, UC-02	SCR-01, 15, 16	TC-001 to 005
Patients	FR-PAT-01 to 10	UC-04, UC-05	SCR-03 to 05	TC-006 to 009
Doctors	FR-DOC-01 to 07	UC-03	SCR-06, 07	TC-010, 014
Appointments	FR-APT-01 to 11	UC-06, UC-07	SCR-08, 09	TC-011 to 013
Consultation	FR-MED-01 to 08	UC-08	SCR-10	TC-015 to 017
Admission	FR-ADM-01 to 09	UC-09, 10, 15	SCR-11	TC-018 to 020
Billing	FR-BIL-01 to 11	UC-11, UC-12	SCR-12, 13	TC-021 to 024
Reports	FR-RPT-01 to 07	UC-13, UC-14	SCR-02, 14	TC-025
Notifications	FR-NOT-01 to 04	UC-06, UC-07	SCR-02	Checked during TC-011, TC-013
Non-functional	NFR-PERF, SEC, USE, REL, MNT, LEG	All	All	TC-026 to 028 and project-wide checks

9.5 Defect severity levels

Severity	Meaning	Example
Critical	System unusable or data loss or security breach; must be fixed before demo	Any user can open the Admin page
Major	A key function fails and there is no workaround	Double booking is allowed
Minor	A function fails but a workaround exists	Filter by status does not reset
Cosmetic	Visual or wording issue only	Misaligned button

10. Project Plan, Roles and Risks

10.1 Suggested development plan

The team should work in short iterations of about two weeks, demonstrating a working increment to the faculty guide at the end of each iteration.

Phase	Weeks	Main activities	Deliverables
1. Requirements and planning	1 to 2	Finalise this SRS, user stories, backlog and task allocation	SRS v1.0, product backlog
2. Design	3 to 4	Wireframes, UI mock-ups, database schema, API design	Figma screens, SQL schema, API list
3. Build, part 1	5 to 6	Login and roles, users, patients, doctors and departments	Working login, patient and doctor modules
4. Build, part 2	7 to 8	Appointments, slots, notifications, consultation records	Booking and consultation flow
5. Build, part 3	9 to 10	Beds and admission, billing and payments, dashboard and reports	End-to-end workflow
6. Testing and fixes	11	System, security and acceptance testing; bug fixing	Test report, defect log
7. Documentation and demo	12	User manual, final report, presentation, demo data	Final submission and demo

10.2 Team roles

Role	Responsibilities	Team member
Project manager and documentation lead	Plans sprints, tracks progress, maintains SRS and final report	[Name]
UI/UX and front-end developer	Wireframes, lime-green theme, HTML, CSS and JavaScript screens	[Name]
Back-end developer	REST API, business rules, authentication and RBAC	[Name]
Database developer	Schema, queries, indexes, backup scripts, sample data	[Name]
Quality assurance lead	Test plan, test cases, defect tracking, acceptance checklist	[Name]

10.3 Risk register

Risk	Likelihood	Impact	Mitigation
Scope grows beyond what the team can finish	High	High	Build High-priority requirements first; freeze scope after Phase 2; keep Low items optional
Team members have uneven technical skills	Medium	Medium	Pair programming; assign modules by strength; short learning tasks in Phase 1

Risk	Likelihood	Impact	Mitigation
Appointment clash logic has bugs	Medium	High	Add a database unique rule and unit tests for slot generation (BR-02)
Security mistakes such as missing role checks	Medium	High	Check roles on the server for every endpoint; use the role matrix in Section 8.3 as a checklist
Data loss during development	Low	High	Use Git, daily database dumps and a shared backup folder
Time lost to integration problems	Medium	Medium	Agree on API contracts early; integrate every week
Late changes in faculty expectations	Medium	Medium	Share the SRS early and record agreed changes in the revision history
Demo fails because of environment issues	Low	High	Prepare a ready demo database and a recorded backup demo video

10.4 Project deliverables

- Software Requirements Specification (this document).
- UI wireframes and mock-ups in the lime-green theme.
- Database schema script and sample data script.
- Source code in a Git repository with a README.
- Test plan, test cases and test report.
- User manual for each role.
- Final report, presentation slides and a working demonstration.

Appendix A. Requirements Summary and Open Issues

A.1 Requirements summary

Category	Count	Identifier range
Functional requirements	77	FR-AUTH, PAT, DOC, APT, MED, ADM, BIL, RPT, NOT
Non-functional requirements	36	NFR-PERF, SEC, USE, REL, MNT, LEG
Use cases	15	UC-01 to UC-15
Screens	16	SCR-01 to SCR-16
Business rules	14	BR-01 to BR-14
Sample test cases	28	TC-001 to TC-028

A.2 Open issues and items to be decided

No.	Issue	Proposed decision	Owner
1	Final technology stack	Confirm Node.js with MySQL or the team's strongest alternative by end of week 2	Project manager
2	Tax percentage for invoices	Use 5% as a configurable default unless the faculty advises otherwise	Project manager
3	Maximum discount without approval	Use 20% as a configurable default	Project manager
4	Patient self-registration on the portal	Not in version 1.0; staff create portal accounts	Whole team
5	Email reminders	Optional; decide the SMTP provider if time permits	Back-end developer
6	Hospital name, logo and address for printouts	Use the working name Carewell with a placeholder logo	UI/UX developer

End of document. Any change to this SRS shall be recorded in the revision history on the document control page and agreed by the team and the faculty guide.